

7-6 Solve and Graph Inequalities

Equation Detective Agency | Standard: 6.EE.5 | Color-by-Number Practice

Learning Goal: I can solve an inequality and graph its solution set on a number line.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: solution set, inequality, inverse operation

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Solve: $x + 3 > 10$. Work: _____ Answer: _____	2. Solve: $x - 5 < 9$. Work: _____ Answer: _____
3. Solve: $2x \geq 12$. Work: _____ Answer: _____	4. Solve: $x/4 \leq 8$. Work: _____ Answer: _____
5. Solve: $x + 6 \leq 15$. Work: _____ Answer: _____	6. Solve: $x - 8 > 1$. Work: _____ Answer: _____
7. Solve: $3x < 21$. Work: _____ Answer: _____	8. Solve: $x/5 \geq 6$. Work: _____ Answer: _____
9. Solve: $x + 10 > 25$. Work: _____ Answer: _____	10. Solve: $x - 4 \leq 11$. Work: _____ Answer: _____
11. Solve: $4x \geq 40$. Work: _____ Answer: _____	12. Solve: $x/6 < 7$. Work: _____ Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

7-6 Mystery Picture: Detective Badge Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
$x > 7 = \text{red}$	$x < 14 = \text{orange}$	$x \geq 6 = \text{yellow}$	$x \leq 32 = \text{green}$
$x \leq 9 = \text{blue}$	$x > 9 = \text{purple}$	$x < 7 = \text{pink}$	$x \geq 30 = \text{brown}$
$x > 15 = \text{gray}$	$x \leq 15 = \text{teal}$	$x \geq 10 = \text{black}$	$x < 42 = \text{light blue}$

2	2	1	2	2	2	3	3	3	4	4	6
2	5	9	5	5	5	6	6	6	7	9	6
1	8	8	9	8	8	9	9	9	12	9	6
3	6	10	2	2	1	2	2	4	2	11	6
3	6	12	2	5	5	5	7	4	2	9	6
3	6	9	1	5	8	11	7	4	12	9	6
5	8	11	11	6	12	1	10	4	2	11	8
1	8	11	3	8	8	10	8	5	2	11	8
5	8	11	5	5	5	4	5	5	3	11	8
7	10	3	4	4	2	3	3	4	5	9	10
7	12	12	1	11	11	12	12	12	2	2	11
9	9	11	8	8	8	9	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	$x > 7$	red	Use inverse operations, then keep the inequality direction for positive numbers.
2	$x < 14$	orange	Use inverse operations, then keep the inequality direction for positive numbers.
3	$x \geq 6$	yellow	Use inverse operations, then keep the inequality direction for positive numbers.
4	$x \leq 32$	green	Use inverse operations, then keep the inequality direction for positive numbers.
5	$x \leq 9$	blue	Use inverse operations, then keep the inequality direction for positive numbers.
6	$x > 9$	purple	Use inverse operations, then keep the inequality direction for positive numbers.
7	$x < 7$	pink	Use inverse operations, then keep the inequality direction for positive numbers.
8	$x \geq 30$	brown	Use inverse operations, then keep the inequality direction for positive numbers.
9	$x > 15$	gray	Use inverse operations, then keep the inequality direction for positive numbers.
10	$x \leq 15$	teal	Use inverse operations, then keep the inequality direction for positive numbers.
11	$x \geq 10$	black	Use inverse operations, then keep the inequality direction for positive numbers.
12	$x < 42$	light blue	Use inverse operations, then keep the inequality direction for positive numbers.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.